

# Public Key Cryptography | RSA Algorithm Example

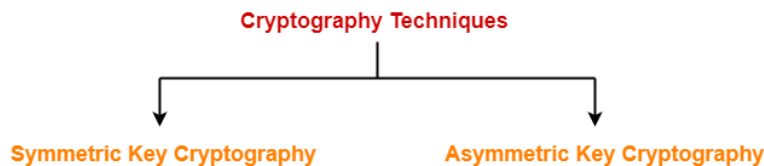
📁 Computer Networks

## Cryptography in Network Security-

Before you go through this article, make sure that you have gone through the previous article on [Cryptography](#).

We have discussed-

- Cryptography is a method of storing and transmitting data in a particular form.
- Cryptography techniques are-



In this article, we will discuss about Asymmetric Key Cryptography.

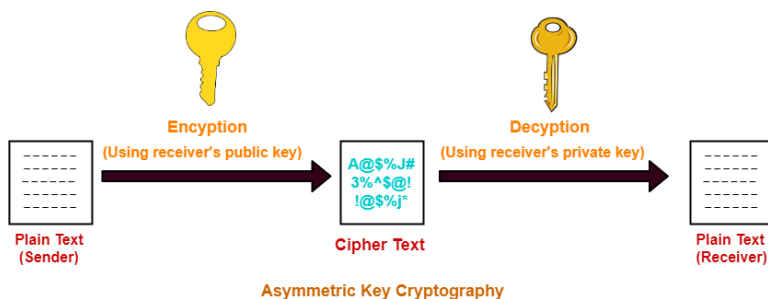
## Asymmetric Key Cryptography-

In this technique,

- Sender and receiver use different keys to encrypt and decrypt the message.
- It is called so because sender and receiver use different keys.
- It is also called as **public key cryptography**.

## Working-

The message exchange using public key cryptography involves the following steps-



## Step-01:

At sender side,

- Sender encrypts the message using receiver's public key.
- The public key of receiver is publicly available and known to everyone.
- Encryption converts the message into a cipher text.
- This cipher text can be decrypted only using the receiver's private key.

## Step-02:

- The cipher text is sent to the receiver over the communication channel.

## Step-03:

At receiver side,

- Receiver decrypts the cipher text using his private key.

- The private key of the receiver is known only to the receiver.
- Using the public key, it is not possible for anyone to determine the receiver's private key.
- After decryption, cipher text converts back into a readable format.

## **Advantages-**

The advantages of public key cryptography are-

- It is more robust.
- It is less susceptible to third-party security breach attempts.

## **Disadvantages-**

The disadvantages of public key cryptography are-

- It involves high computational requirements.
- It is slower than symmetric key cryptography.

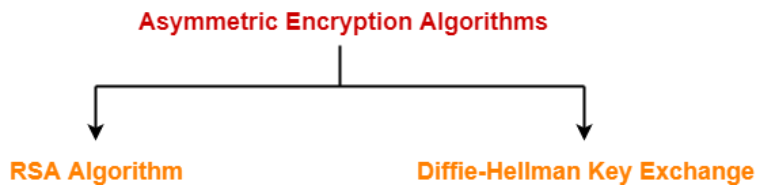
## **Number of Keys Required-**

To use public key cryptography,

- Each individual requires two keys- one public key and one private key.
- For  $n$  individuals to communicate, number of keys required =  $2 \times n = 2n$  keys.

## **Asymmetric Encryption Algorithms-**

The famous asymmetric encryption algorithms are-



1. RSA Algorithm
2. Diffie-Hellman Key Exchange

In this article, we will discuss about RSA Algorithm.

## **RSA Algorithm-**

Let-

- Public key of the receiver =  $(e, n)$
- Private key of the receiver =  $(d, n)$

Then, RSA Algorithm works in the following steps-

### **Step-01:**

At sender side,

- Sender represents the message to be sent as an integer between 0 and  $n-1$ .
- Sender encrypts the message using the public key of receiver.
- It raises the plain text message 'P' to the  $e^{\text{th}}$  power modulo  $n$ .
- This converts the message into cipher text 'C'.

$$C = P^e \bmod n$$

## **Step-02:**

- The cipher text 'C' is sent to the receiver over the communication channel.

## **Step-03:**

At receiver side,

- Receiver decrypts the cipher text using his private key.
- It raises the cipher text 'C' to the  $d^{\text{th}}$  power modulo  $n$ .
- This converts the cipher text back into the plain text 'P'.

$$P = C^d \bmod n$$

## **NOTE-**

**'e' and 'd' must be multiplicative inverses modulo  $\phi(n)$**

After decryption, receiver must have-

$$P = C^d \bmod n$$

$$P = (P^e \bmod n)^d \bmod n$$

$$P = P^{ed} \bmod n$$

For this equation to be true, by Euler's Theorem, we must have-

$$ed = 1 \bmod \phi(n)$$

OR

$$ed = k\phi(n) + 1$$

Thus, e and d must be multiplicative inverses modulo  $\phi(n)$ .

## **Steps to Generate Public Key And Private Key-**

An individual can generate his public key and private key using the following steps-

### **Step-01:**

Choose any two prime numbers p and q such that-

- They are different.
- They are very large.

### **Step-02:**

Calculate 'n' and totient function  $\phi(n)$  where-

- $n = p \times q$
- $\phi(n) = (p-1) \times (q-1)$

### **Step-03:**

Choose any value of 'e' such that-

- $1 < e < \phi(n)$
- $\gcd(e, \phi(n)) = 1$

### **Step-04:**

Determine 'd' such that-

$$ed = 1 \bmod \phi(n)$$

OR

$$d = \frac{1 + k \phi(n)}{e}$$

- You already know the value of 'e' and  $\phi(n)$ .
- Choose the least positive integer value of 'k' which gives the integer value of 'd' as a result.
- Use trial and error method.
- Start substituting different values of 'k' from 0.

## **PRACTICE PROBLEMS BASED ON RSA ALGORITHM-**

### **Problem-01:**

In a RSA cryptosystem, a participant A uses two prime numbers  $p = 13$  and  $q = 17$  to generate her public and private keys. If the public key of A is 35, then the private key of A is \_\_\_\_\_.

## **Solution-**

Given-

- Prime numbers  $p = 13$  and  $q = 17$
- Public key = 35

### **Step-01:**

Calculate 'n' and totient function  $\phi(n)$ .

Value of n,

$$n = p \times q$$

$$n = 13 \times 17$$

$$\therefore n = 221$$

Totient function,

$$\phi(n) = (p-1) \times (q-1)$$

$$\phi(n) = (13-1) \times (17-1)$$

$$\therefore \phi(n) = 192$$

### **Step-02:**

- We are already given the value of  $e = 35$ .
- Thus, public key =  $(e, n) = (35, 221)$

### **Step-03:**

Determine 'd' such that-



$$d = \frac{1 + k \phi(n)}{e}$$

$$d = \frac{1 + k \times 192}{35}$$

Here,

- The least value of 'k' which gives the integer value of 'd' is  $k = 2$ .
- On substituting  $k = 2$ , we get  $d = 11$ .

Thus, private key of participant A =  $(d, n) = (11, 221)$ .

## **Problem-02:**

In the RSA public key cryptosystem, the private and public keys are  $(e, n)$  and  $(d, n)$  respectively, where  $n = p \times q$  and  $p$  and  $q$  are large primes. Besides,  $n$  is public and  $p$  and  $q$  are private. Let  $M$  be an integer such that  $0 < M < n$  and  $f(n) = (p-1)(q-1)$ .

Now consider the following equations-

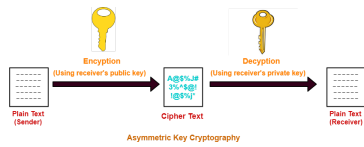
I.  $M' = M^e \bmod n$  and  $M = (M')^d \bmod n$

II.  $ed \equiv 1 \bmod n$

III.  $ed = 1 \bmod f(n)$

IV.  $M' = M^e \bmod f(n)$  and  $M = (M')^d \bmod f(n)$

1. I and II
2. I and III
3. II and IV
4. III and IV



**Article Name** Public Key Cryptography | RSA Algorithm Example

**Description** Public key cryptography or Asymmetric key cryptography use different keys for encryption and decryption. RSA Algorithm Examples. RSA Algorithm and Diffie Hellman Key Exchange are asymmetric key algorithms.

**Author** Akshay Singhal

**Publisher Name** Gate Vidyalay

**Publisher Logo**





|                                                          |                                                                        |                                                        |                                           |
|----------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------|
| <b>Checksum in<br/>Networking  <br/>Checksum Example</b> | <b>Diffie Hellman Key<br/>Exchange  <br/>Asymmetric<br/>Encryption</b> | <b>Cyclic Redundancy<br/>Check   CRC  <br/>Example</b> | <b>Dijkstra-Algorithm-<br/>Problem-01</b> |
|----------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------|

|                                    |                                                                    |                                                                                      |                                      |
|------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------|
| <b>Computer Networks<br/>Notes</b> | <b>Error Detection in<br/>Computer Networks<br/>  Parity Check</b> | <b>Dijkstra-Algorithm-<br/>Problem-01-<br/>Solution-Step-04-<br/>Before-Edge-...</b> | <b>Dijk<br/>Pro<br/>Sol<br/>Afte</b> |
|------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------|

---